

Linux Programming: Assignment-5:

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CLASS- CYBER SECURITY

SECTION- C

1.What is a shell in Linux OS? How many categories of shell is currently exists in Linux? Why bash shell is very popular in Linux distribution?

Ans:

Shell in linux is a command line interface thats connects user and operating system kernel it also allows user to enter commanda which shell interprets and executes to perform tasks for example file management and process control

Categories of shell in linux:

- 1.Bourne shell
- 2.C Shell
- 3.Korn Shell (ksh)
- 4.Bourne Again Shell (bash)
- 5.Z Shell (zsh)
- 6.Fish Shell (fish)

Bash is popular because its has features like command history,tab completion and scripting capabilities it is also user

friendly easy to learn and document. Bash is highly customizable and open source which makes it easy for people to port their work.

2. What does the ls -Z command display?

Ans:

ls -Z command is used to display the security context of each file that system has, it shows details like user,role type level

Output might look like

```
-rw-r--r--. root root system_u:object_r:etc_t:s0 config.txt
```

3. Write a command to list all hidden files in the current directory.

Ans:

To list all hidden files in directory we can use ls -a command which displays all the hidden files the 'a' stands for all

In linux hidden files are those which start with a dot operator

4. Explain the difference between hard links and soft links (symbolic links) in Linux

Ans:

Hard links:

Its the direct reference to same data on disk as original files

In hard link both original file and hard link share the same inode number

If original file is deleted then the data exists as long as the one hard link remains

Command: ln original.txt hardlink.txt

Soft link:

Soft link is like a shortcut to another file

It has different inode number than original file but does point to the original file's path

If original file is deleted then the soft link becomes broken

Command: `ln -s original.txt softlink.txt`

5. A file has permissions -rwxr-x--x. Explain who can read, write, and execute it.

Ans:

The rwx part in the command lets the owner or user to read write and execute the file and r-x part lets a group read and execute only, the -- x part lets other to execute only

6. Write the command to change the group ownership of a file data.txt to group staff.

Ans:

Command to use: `chgrp staff data.txt`

7. Why is it dangerous to give 777 permissions to a file? Explain with an example.

Ans:

777 permission means that everyone (user, group, and others) has read write and execute access to it. It is dangerous because it can be a security risk as anyone can modify, delete, or execute the file.

Data can be lost as a user can easily get access and accidentally or intentionally delete or overwrite the file.

If the file is an executable or script, other could insert harmful code and run hence it is dangerous to give a file 777 permission.

8. What is the difference between apropos (i.e., man -k) and whatis (i.e., man -f)?

Ans:

Apropos:

This command searches the man page database for keyword in command description

This command is used when user dont know the exact command name but know what they are looking for

Whatis:

This command displays a small summary or one line deatils about the specific command

This command is used when you already know the command name

9. Write a command to redirect the error output of a command to a file named error.log.

Command:

command 2> error.log

2> → redirects **standard error (stderr)**.

error.log → the file where the **error messages** will be saved.

command → any Linux command you run.

10. How can you use the tee command to append output to a file instead of overwriting it?

Ans:

To append the output we can use the command:

command | tee -a filename

Here:

'|' sends the output of command to another command

'tee' writes output both to file terminal and to a file

'-a' appends the output